

Topology 4.1

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Problem 4.1 - 1

- a) Let X have the discrete topology and Y be an arbitrary topological space. Then every function $f : X \rightarrow Y$ is continuous.

Proof. .

1. Let U be an open set in Y .
2. Then $\forall f(x) \in U, \{x\}$ is open in X .
3. Since the union of open sets is open, then $f^{-1}(U)$ is open.

□

- b) Let Y have the trivial topology and X be an arbitrary topological space. Then every function $f : X \rightarrow Y$ is continuous.

Proof. .

1. Observe that the only topologies on Y are $\emptyset$ and Y .
2. Then $f^{-1}(\emptyset) = \emptyset$ which is open in X , and $f^{-1}(Y) = X$ (since f is a function) which is open in X .

□

Problem 4.1 - 2

Let X and Y be topological spaces. A function $f : X \rightarrow Y$ is continuous iff $f^{-1}(C)$ is closed in X for every closed set C in Y .

Proof. .

1. ($\Rightarrow$) Let f be a continuous function, and let C be closed in Y .
2. Then $Y - C$ is open in Y , and so $f^{-1}(Y - C)$ is open in X .
3. Observe that $f^{-1}(Y - C) = f^{-1}(Y) - f^{-1}(C) = X - f^{-1}(C)$, so $f^{-1}(C)$ is closed in X .
4. ($\Leftarrow$) Assume that $f^{-1}(C)$ is closed in X for every closed set C in Y .
5. Let $U = Y - C$ be some open set in Y where C is some closed set in Y .
6. Then $f^{-1}(U) = f^{-1}(Y - C) = f^{-1}(Y) - f^{-1}(C) = X - f^{-1}(C)$
7. Since $f^{-1}(C)$ is closed in X , then $X - f^{-1}(C)$ is open in X .
8. Therefore, for all open sets U in Y , $f^{-1}(U)$ is open in X and so f is continuous.

□

Problem 4.1 - 3

The function $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous in $\varepsilon - \delta$ definition iff $\forall x \in \mathbb{R}$ and $\forall U$ where $f(x) \in U$, $\exists V$ where $x \in V$ such that $f(V) \subset U$.

Proof. .

1. ($\Rightarrow$) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous in $\varepsilon - \delta$ definition, and let $f(x) \in U$ where $x \in \mathbb{R}$.
2. Then there must exist some basis element $B \subset U$ such that $f(x) \in B = (a, b)$.
3. Let $\varepsilon = \min\{|a - f(x)|, |b - f(x)|\}$, and observe that $(f(x) - \varepsilon, f(x) + \varepsilon) \subset B \subset U$.
4. Then by the $\varepsilon - \delta$ definition, $\exists \delta$ such that whenever $x \in (x - \delta, x + \delta) = V$, then $f(x) \in (f(x) - \varepsilon, f(x) + \varepsilon) = f(V) \subset U$.
5. Then $\forall U$ where $f(x) \in U$, $\exists V$ where $x \in V$ such that $f(V) \subset U$.
6. ($\Leftarrow$) Assume that for all neighborhoods U of $f(x)$, there exists some neighborhood V of x such that $f(V) \subset U$.
7. Let $x \in \mathbb{R}$, let $\varepsilon > 0$, and let $U_\varepsilon = (f(x) - \varepsilon, f(x) + \varepsilon)$.
8. Observe that $f(x) \in U_\varepsilon$.
9. Then there must exist some open set V containing x such that $f(V) \subset U_\varepsilon$.
10. Since V is open, there must be some basis element $(a, b) \subset V$ such that $x \in (a, b)$.
11. Let $\delta = \min\{|a - x|, |b - x|\}$, and observe that $x \in (x - \delta, x + \delta) = V_\delta \subset V$.
12. Then $f(V_\delta) \subset U_\varepsilon$.
13. Then for any x and any $\varepsilon > 0$, $\exists \delta$ such that $x \in (x - \delta, x + \delta)$ implies $f(x) \in (x - \delta, x + \delta)$.
14. Therefore f is continuous in the $\varepsilon - \delta$ definition.

□

Problem 4.1 - 6

- a) Let $p_X : X \times Y \rightarrow X$ and $p_Y : X \times Y \rightarrow Y$ where $p_X(x, y) = x$ and $p_Y(x, y) = y$. Then p_X and p_Y are continuous.

Proof. .

1. First note that p_X and p_Y are functions, and therefore must be defined on the entire domain. This means that for any $x \in X$, then $p_X^{-1}(x) = \{x\} \times Y$, and similarly for any $y \in Y$ then $p_Y^{-1}(y) = X \times \{y\}$.
2. Let U be an open set in X .
3. Then $p_X^{-1}(U) = U \times Y$.
4. Observe that U is open in X and Y is open in Y .
5. Therefore $U \times Y$ is open in $X \times Y$, and so p_X is a continuous function.
6. Now, let V be open in Y , and observe that $p_Y^{-1}(V) = X \times V$ which is open in $X \times Y$.
7. Therefore p_Y is also continuous.

□

- b) The Product topology on $X \times Y$ is the coarsest topology for which p_X and p_Y are continuous.

Proof. .

1. Let T be the product topology on $X \times Y$, and let T' be some topology such that p_X and p_Y are continuous.
2. We will show that $T \subset T'$, and therefore more coarse than T' .
3. Let $U_X \times U_Y$ be some open set in T and observe that U_X is open in X and U_Y is open in Y .
4. Since p_X is continuous, then $p_X^{-1}(U_X) = U_X \times Y$ must be open in T' .
5. Since p_Y is continuous, then $p_Y^{-1}(U_Y) = X \times U_Y$ must be open in T' .
6. Since both $U_X \times Y$ and $X \times U_Y$ are open, then $U_X \times Y \cap X \times U_Y = U_X \times U_Y$ is open in T' .
7. Therefore $T \subset T'$.

□

Problem 4.1 - 7

Let $id : X_{T_1} \rightarrow X_{T_2}$ where $id(x) = x$. Then id is continuous iff $T_2 \subset T_1$.

Proof. .

1. $(\Rightarrow)$ Assume that id is a continuous function.
2. Let U be some open set in T_2 .
3. Observe that $id^{-1}(U) = U$, and so U must be open in T_1 .
4. Therefore $T_2 \subset T_1$.
5. $(\Leftarrow)$ Assume $T_2 \subset T_1$.
6. Let U be an open set in T_2 .
7. Observe that $id^{-1}(U) = U$ which is open in T_1 since $T_2 \subset T_1$.
8. Therefore id is continuous.

□

Problem 4.1 - 10

Let $f : X \rightarrow Y$ be a function with a graph $G = \{(x, f(x)) | x \in X\}$ as a subset of $X \times Y$. If f is continuous and Y is Hausdorff, then G is closed.

Proof. .

1. Assume f is continuous and Y is Hausdorff.
2. Let $(x, y) \in (X \times Y) - G$.
3. Then $y \neq f(x)$, and so $\exists U, V$ as open sets in Y such that $y \in U$, $f(x) \in V$, and $U \cap V = \emptyset$ since Y is Hausdorff.
4. Since f is continuous, $\exists W$ as an open set in X where $x \in W$, such that $f(W) \subset V$.
5. Then $U \times W$ is disjoint from G and is open in $X \times Y$.
6. Then for every point $(x, y) \in (X \times Y) - G$, we can find an open set $U \times W$ containing (x, y) such that $U \times W \subset (X \times Y) - G$.
7. Therefore $(X \times Y) - G$ is an open set, and so G is closed.

□

Problem 4.1 - 11

Let $f : X \rightarrow Y$ be continuous, and let A be a subspace of X . Then $f|_A : A \rightarrow Y$ is continuous.

Proof. .

1. Let U be an open set in Y .
2. Since f is continuous, $f^{-1}(U)$ is open in X .
3. Then $f|_A^{-1}(U) = f^{-1}(U) \cap A$, which is open in A .
4. Therefore $f|_A$ is continuous.

□